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MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Multi-Disciplinary Courses (MDC) - MG1MDCZGY100 - ORNAMENTAL FISH FARMING AND AQUARIUM KEEPING
(2024 ADMISSION ONWARDS)

Duration: 1 hour **Maximum Marks: 35**

Part A

Multiple Choice Questions (Answer All Questions. Each Question carries 1 mark)

1. ……. is an aquarium plant. **[U] / [CO4]**

Java fern, Vallisneria, *Anubias*, Amazon sword, Hornwort, *Rotala*, Java Moss
(Any one aquarium plant – give full marks)

2. The mechanism of swallowing the juveniles of the same species is …….. [An] / [CO1]

Cannibalism (filial cannibalism)

3. The common name for *Xiphophorus hellerii* is _____. [K] / [CO1]

Swordtail

4. Excess buildup of fish excreta may lead to the release of ---------- gas in aquariums. **[U] / [CO4]**
Ammonia

5. The least expensive method of tranquillization during fish transport is …..... A] / [CO2]

Use of cold water (5-10ºC)

(5 x 1 = 5)

Part B

Short Answer Type Questions (Answer any 5 Questions. Each Question carries 2 marks)

6. List the macronutrients needed by fish for survival. **[U] / [CO1]**

The primary macronutrients needed by fish for survival are **proteins**, **carbohydrates**, and **lipids** (fats).

7. Write a short note on community aquaria. **[A] / [CO3, CO4]**

Community aquaria are tanks that are **designed to contain more than one species of fish.** They include a variety of species that do not normally occur together in nature, for example, angelfish, swordtails, and gouramis. Such communities aim to bring together **fish that are compatible in temperament and water requirements**, while using their different colours and behaviours to add interest and entertainment value.

8. Give an account of a method of growing microscopic organisms in a medium to feed fish fry and larvae. **[U] / [CO1, CO4]**

i. **Algae**: minute floating, aquatic plant-like organisms.
Groundnut oil cake, urea, and superphosphate are mixed and diluted in water and added to cement tanks along with algae from a stock culture. The algae are collected and fed to the fish larvae.

ii. **Rotifers**: free-swimming minute organisms in seawater known as 'wheel animalcules'.
Filtered seawater, urea and superphosphate are added to the water, along with the stock culture of chlorella. A stock culture of rotifers is added to this mixture on the next day. Salinity (30ppt) and temperature (25-35°C). In 6-7 days, rotifers multiply and increase in concentration.

iii. **Moina**: free-floating minute organisms belonging to a group of animals known as cladocerans.
Filtered seawater, urea and superphosphate are added to the water, along with the stock culture of chlorella. Moina stock culture is added to it. In 5-7 days, Moina cell density increases.

iv. **Daphnia**: floating animals belong to the cladocerans.
Pond water is manured with chicken droppings, oil cakes and superphosphate, and in a few days the water turns green, indicating growth of plankton. Daphnia is introduced to it. Liquid manure is added to the tank daily.

v. **Infusoria**: the smallest of all live feeds for fish larvae
Water is taken in a tank, and banana peel, cabbage leaves or potato pieces are added and left in the shade. Medium for infusoria can also be developed by using aquatic plants, skimmed milk, etc. The medium is covered by a velon screen and kept for bacterial development. Decaying vegetation acts as food for the bacteria; the infusoria feed on these bacteria and multiply.

vi. **Earthworm**: found in moist and loose soils. They are washed to remove the adhering soil and killed by pouring hot water or kept in live conditions.

vii. **Mosquito larvae**: collected from stagnant pools of dirty water and cultured. Take water in a bucket, add cow dung or decaying leaves and keep undisturbed in a place away from sunlight and fresh air. Mosquitoes lay eggs in the water; the larvae can be collected by a sieve and given to fish.

viii. **Artemia**: also known as brine shrimp, inhabit salt pans of high salinity.
The cysts are put in saline water (1-5 g/l) at a temperature of 27-30°C and aerated for 24-26 hours. When they hatch out, aeration can be stopped and the larvae can be attracted by a light source and siphoned off from the jar, washed and given to the fish as live feed.

(Any 2 points with short description)

9. Describe the breeding habits of Gourami. [U] / [CO1]

Gouramis are one of the popular aquarium species, being **bubble-nesters**, where the male builds a nest of bubbles for eggs, which he then guards**. During courtship, the male and female care for each other**; the **male nudges the female into a position beneath the nest.** The male curves his body around the female and tips her over so that her abdomen is directed towards the nest. Spawning lasts for several hours. **Eggs are light and float upon release**. The male carefully gathers any stragglers and places them back in the nest. **Once the fry hatches, the male should be removed.**

(Liberal valuation)

10. Discuss the importance of water quality monitoring in feeding management. [K] / [CO1]

Water quality monitoring is crucial for feeding management because it directly impacts **animal health, disease resistance, feed efficiency, and the quality of the aquatic life**. By regularly monitoring parameters like ***dissolved oxygen, pH, temperature, hardness, turbidity, salinity, and alkalinity***, aquarium keepers can prevent disease, optimise feeding strategies, and ensure that the water used in processing doesn't contaminate the food. This proactive approach is essential for maximising productivity and minimising costs associated with poor water quality.

11. Write a short note on Hang-On-Back filters. **[A] / [CO4]**

Hang-On-Back filters are a type of **mechanical filter** that is mounted on the back of the aquarium. It includes a **filter cartridge** that contains a foam or polyester pad **to capture debris**, which is easy to install and maintain.

(Liberal valuation)

12. Write notes on larval rearing of Gourami. **[U] / [CO3]**

The eggs of gourami hatch out within a day, and the fry remain in the floating bubble nest. **Infusoria** is given as **live feed**, supplemented with fine freeze-dried food or finely smashed boiled egg yolk. The **male fish is removed from the tank** during this time. The tank is kept covered to prevent any sudden change in the water temperature. If properly fed, the young ones become sexually mature within a year.

(Give full marks if any relevant points written)

(5 x 2 = 10)

Part C

Short Essay Type Questions (Answer any 5 Questions. Each Question carries 4 marks)

13. Discuss briefly any four diseases in aquarium fishes caused by parasites. **[U] / [CO3]**

Parasites are organisms that derive sustenance from their host while causing it harm. There are three major groups of parasites that infect fish: protistan, helminth and parasitic arthropods.

1. **Ich disease (White spot disease or Ichthyophthiriasis)** - *Ichthyophthirius*
   Symptoms: Presence of small, white, salt-like cysts on the skin, fins, and gills, excessive mucus on the body, rapid gill movement, and difficulty in breathing.
2. **Costiasis (Ichthyobodiasis)** - *Costia necatrix*
   Symptoms: Red patches appear on the skin, gills turn brown and are partially destroyed, increased mucus production, and appetite loss.
3. **Gyrodactylosis** - *Gyrodactylus*
   Symptoms: It is seen on the body, fins and gills of fish and causes local irritation, increased mucus production, and damages the epithelial tissues, and the colour of the fish fades.
4. **Dactylogyrosis** - *Dactylogyrus*
   Symptoms: Infected fish display sores and ulcers on the skin and inflamed gills, have a pale skin colour, and are covered with small haemorrhages.
5. **Argulosis** - *Argulus*
   Symptoms: Found attached to the skin and fins of fish and feed on the body fluids. Reddish lesions occur at the site of attachment, and this opens up the skin to secondary bacterial and fungal infections. Anaemia and stunted growth occur.
6. **Lernaeasis** - *Lernaea*
   Symptoms: Found attached to the skin, gills and fins by means of its hooks and feeds on the body fluids. The fish shows poor appetite, anaemia, stunted growth, and reddish spots occur at the site of attachment.

(Mention any four diseases and brief explanation – 1 mark each)

14. Define induced breeding. Add notes on the methods of Induced breeding in fish. **[U] / [CO3]**

Fishes that do not spawn in natural conditions **are made to spawn and lay eggs by artificial methods** is called induced breeding. Mainly done by two methods:

1. **Hypophysation**: By injecting pituitary gland extract, which contains hormones called gonadotropins - ***Follicle Stimulating Hormone (FSH)*** *and* ***Luteinizing Hormone (LH)***, which stimulate the fish to spawn.
2. **Injection of Ovaprim**: Ovaprim is a drug used in induced breeding as a substitute for pituitary extract, made up of two components: a ***synthetic gonadotropin-releasing hormone*** (stimulates the sex organs of the fish) and a drug called ***domperidone*** (inhibits the action of dopamine, a substance produced by the fish that inhibits ovulation).

(Definition – 1 mark; Methods – 1.5 marks each)

15. Write an account on the importance of water quality and environmental stability in the breeding success of bettas, and discuss best practices for maintaining these conditions. [A] / [CO3]

The Siamese fighting fish, *Betta splendens*, are known for their brilliant colours, large flowing fins and their peculiar territorial behaviour. They exhibit parental care and bubble nest-building behaviour. Bettas are sensitive to fluctuations in their environment, and stable water parameters are essential for stimulating spawning behaviour and ensuring the survival of eggs and fry. For broodstock development, the **males and females are grown in separate tanks** upon attaining maturity. Broodstock is then taken to **50L breeding tanks, spacious enough** for the males to chase the females. The **bottom of the tank is kept bare to enable the males to pick up loose eggs and stick them onto the nest.** The **temperature** is kept at **20-25°C** and **pH** at **7-7.5**. A few **pieces of plants (like banana leaves or floating plants) may be kept floating in water to provide adequate hiding spots for the female and for the egg mass to be attached by the male.** Bettas **prefer calm waters, so care to be taken to use a filter with an adjustable flow and set it to a gentle output. Stable lighting is important for regulating their natural rhythms and breeding behaviour.** The breeding tank is partitioned into two; in one half, the male is kept, while the females are kept in the other half. Males are introduced into the tank first, followed by the females in the other half. The readiness of the female to spawn is indicated by vertical stripes on the body and white spots near the anus. Males start building a bubble nest, producing bubbles coated with saliva and stuck together, floating on the water surface. The partition is removed a few hours later, and females are allowed to mix with the males. At this stage, the males should accept the females; otherwise, the female is chased away by the male through aggressive displays. Fighters often spawn early in the morning. The females are guided to a position below the nest; the male curls his body spirally around the female so that the vents are close together. The female releases eggs, immediately fertilised by the male.

(Liberal valuation)

16. Compare and contrast moist feeds and wet or paste feeds. In what scenarios might each be more beneficial? [E] / [CO1]

Feature	**Moist Feeds**	**Wet or Paste Feeds**
Moisture Content	Typically 25-45%.	Typically 45-70%
Ingredients	A mix of high-moisture ingredients and dry pulverised ingredients.	Primarily made from high-moisture ingredients
Storage & Stability	Better stability than wet feeds, but still requires refrigeration/special storage to prevent spoilage.	Highly perishable; must be stored under refrigeration, or consumed immediately after preparation.

Benefits: **Moist feeds** are more beneficial as they **offer the palatability** and **hydration benefits** of high-moisture feeds with slightly **better handling and storage properties** than wet feeds prepared daily. **Wet feeds** are more beneficial as the **soft texture is easy to consume, making it ideal for fry.**

(Liberal valuation)

17. How do guppies, mollies, and swordtails differ in their approaches to caring for their fry?

[K] / [CO3]

***Guppies*: Provide **no parental care** after birth, and the fry must fend for themselves to find food and avoid predators. **Adults may eat their young**, so separation is recommended. Fry requires special care, including **feeding with appropriately sized food** such as infusoria, baby brine shrimp, or finely crushed flakes**. Water quality must be carefully maintained** to ensure their survival.

***Mollies*: Do not provide parental care**. They **will eat their young**, so it is essential to separate them. Mollies give birth to release the babies. **A varied diet of live and flake foods is a good option for feeding.** Live food includes baby brine shrimp, microworms, *Daphnia*, and hard-boiled egg yolk paste can be given as a protein-rich alternative for live feeds. **Fry requires frequent feeding** (5 times a day.)

***Swordtails*: Do not provide parental care. Adult** swordtails will **eat their young**, making separation necessary to protect them. If it is not possible to transfer the parents or fry from the tank after breeding, **a lot of plants like Java Moss are provided for the fry to hide** themselves from the parents. The fry can be **fed with artificial feed in the form of flakes**; **newly hatched brine shrimp provides a more wholesome live feed.** The combination of *Spirulina* algae and brine shrimp is known to make the fry grow very rapidly and display their true colours earlier.

18. Give an account of the sex identification features and broodstock maintenance of swordtail fish. [S] / [CO3]

Swordtail is one of the most popular aquarium fish of the family Poeciliidae. Swordtails are mild in behaviour and can be housed in a community aquarium.

Sex identification: Adult males and females can be easily differentiated. **Females are larger** in size **than the males** and have much more rounded bellies. **Males are thinner**, elongated and develop a **distinct sword-like elongation of the lower lobe of their tail fin** as they mature, but **females do not have an elongated lower lobe** for the caudal fin. **Mature males** exhibit a distinct male sex organ called a **gonopodium** (modification of the anal fin), which is used as a copulatory organ **for depositing sperm into the female during mating.**

Maintenance of broodstock: Swordtail should be **provided with plenty of room to swim** and may require one square foot per individual fish. The tank bottom should contain a substrate of fine sand or smooth gravel. They **prefer a warmer temperature range** of **21-27°C**, with 24°C as the optimum. The **pH** should be maintained at **7-8.4**. **Floating plants should be provided in the tank for the young to hide**, as they exhibit cannibalism. The **water has to be kept well aerated** and heavily planted to prevent the fish from jumping out. Males and females of desirable body shape and colour are selected as the broodstock. **Swordtail can be kept in sets of 3-4 females per male. Minimum feeding is required** as they feed on detritus and algae. Frozen foods like brine shrimps, krill, Daphnia, and plankton can also be given.

(Sex identification features – 2 marks; Broodstock maintenance – 2 marks)

19. Illustrate the role of the nitrogen cycle in an aquarium. [A] / [CO3, CO4]

The nitrogen cycle is a biological process that **converts harmful waste products in an aquarium into less toxic forms**. It involves the breakdown of fish waste, uneaten food, and plant debris into **ammonia**, which is highly toxic to fish. **Beneficial bacteria then convert ammonia into nitrites**, and eventually **into nitrates**, which are much less harmful. **Nitrates** can be **controlled through regular water changes.** This cycle is essential because it ensures that the aquarium environment remains safe for fish by **preventing the buildup of toxic chemicals**. Establishing this cycle before adding fish, known as **tank cycling, is critical for maintaining water quality and fish health.**

(5 x 4 =20)